

Name

- For the ODE $y' = A y$ with A below. Write the gen sol and the important part as $t \rightarrow \infty$.

In[*]:= **MatrixForm[Eigensystem[$\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$]]]**

Out[*]//MatrixForm=

$\begin{pmatrix} 2 & 2 \\ \{1, 0\} & \{0, 0\} \end{pmatrix}$

$y(t) =$	$\approx$	
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1. For the ODE $y' = A y$ with A below. Write the gen sol.

$y(t) =$

In[*]:= **MatrixForm[Eigensystem[$\begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}$]]]**

Out[*]//MatrixForm=

$\begin{pmatrix} 1 & 1 \\ \{1, 1\} & \{0, 0\} \end{pmatrix}$

- For the ODE $y' = A y$ with A below. Write the gen sol and the important part as $t \rightarrow \infty$.

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In[*]:= MatrixForm[Eigensystem[ $\begin{pmatrix} 0 & -5 & -2 \\ 1 & 0 & 1 \\ -2 & 4 & 0 \end{pmatrix}$ ]]
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Out[*]//MatrixForm=
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$$\begin{pmatrix} 2 & -1 & -1 \\ \{-1, 0, 1\} & \{-1, -1, 2\} & \{0, 0, 0\} \end{pmatrix}$$

$y(t) =$	$\approx$	
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2. For the ODE $y' = A y$ with A below. Write the gen sol and the important part as $t \rightarrow \infty$.

$y(t) =$	$\approx$	
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In[1]:= MatrixForm[Eigensystem[ $\begin{pmatrix} 2 & 6 & -3 \\ -2 & -5 & 2 \\ -1 & -2 & -4 \end{pmatrix}$ ]]
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Out[1]//MatrixForm=
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$$\begin{pmatrix} -3 & -3 & -1 \\ \{3, -2, 1\} & \{0, 0, 0\} & \{-2, 1, 0\} \end{pmatrix}$$